M⋆: Every Task Deserves Its Own Memory Harness

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HarnessRSI 90 ★★★

evidence: improver

Paper

Code

1 What changes, what is kept

One memory program = one Python module (all 3 mutate)

Schema	data formats stored / queried	**Logic**	code write() / read()
Instruction	prompts extract/query/use	**Toolkit**	allowed tools lists, DBs, LLM

1 Program pool
3 seeds, then every child; parent drawn by softmax on validation score (τ = 0.15)

↓

2 Evaluate the parent
static validation set gives score J; a rotating batch of 5 items gives traces

↓

3 Coding agent mutates
reads traces + earlier change logs, writes a diff patch (GPT-5.3-Codex)

↓

4 Quality checks
compile, smoke test, ≤ 3000 chars, ≤ 60 s; failure → auto-fix up to 3×, else discard

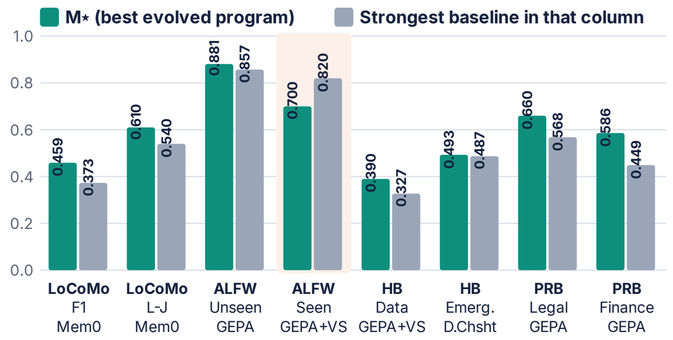
↓

5 Score child, add to pool
child's static-validation score J decides how often it is sampled later

20×

after 20 iterations: best on validation → one test run
weights never change; the kept artifact is code

2 Headline: best on 7 of 8 test metrics



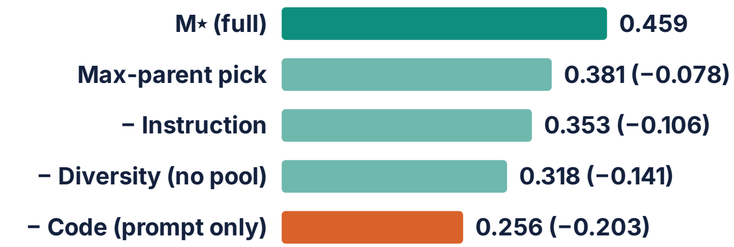
Test split, GPT-5.4 Mini task agent, 20 iterations; Table 1 matches the seed-0 column of Table 13. Test sets: 50–100 items. Shaded: ALFWorld Seen.

4 Where the gain is absent or thin

- **ALFWorld Seen: 0.700** — below GEPA+VS 0.820, Trajectory Retrieval 0.780, GEPA and Vector Search 0.720
- HB Emergency 0.493 vs 0.487; ALFWorld Unseen 0.881 vs 0.857 — margins on 50–100 test items (Unseen: one episode), no CIs
- LoCoMo **temporal 0.19** vs GEPA+VS 0.38; causal 0.35 vs 0.39; single-hop tied with Mem0 (0.33). Strongest on open-domain (0.51)
- **Cost:** $12–90 per 20-iteration run (ALFWorld ≈ 100 h) vs $0.31–$1.51 for No Memory / Vector Search; no matched-compute baseline, no baseline variance

3 Controls that decide if the gain is real

Ablation · LoCoMo F1 (Table 2, single runs)



Seeds · only 3 of 7 configs (Table 3)

5 seeds	mean ± sd	best base	above
LoCoMo F1	0.421 ± 0.042	0.373	**4 / 5**
HB Data	0.391 ± 0.023	0.327	5 / 5
PRB Finance	0.561 ± 0.032	0.449	5 / 5

Table 1 shows seed 0 (0.459 / 0.390 / 0.586); five-seed means are lower on LoCoMo and PRB Finance.

Transfer · program evolved on another task (Fig. 5)

Target task	seed	native	evolved elsewhere
LoCoMo F1	0.26	**0.46**	**0.02 · 0.09**
ALFWorld	0.64	**0.88**	**0.62 · 0.58**
PRBench	0.47	**0.66**	0.06 · **0.58**

5 of 6 transferred programs fall below the universal seed.

Takeaway

Rewriting schema, retrieval code and prompts together, over a population, beats fixed memory harnesses on most tests — not all. The direction holds across seeds, but the size is smaller than the headline, programs mostly fail to transfer, and the search costs $12–90 per run.

Our read · HarnessRSI

Score 90 · ★★★ · evidence level: improver

Reports: held-out test, transfer, 5 seeds (3 configs), cost, ablations.
Does not report: matched-compute baselines, leakage check, baseline variance.